

Meriem Trabelsi

Software Engineer • systems integration and process automation

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Open to relocation within the EU • Arabic (native) • French (C1) • English (B2)

PROFILE

I build systems where the business process is an executed model rather than logic buried in code. My field is the integration of telecom systems aligned with TM Forum standards: BPMN orchestration, event-driven choreography and externalised business rules.

EXPERIENCE

Process Automation Engineer — IT Serv

January 2025 – June 2026 • final-year engineering project, highest honours

Sole designer and developer of an Order-to-Activate orchestration platform for a fibre operator, from address qualification through to the first invoice. A proof of concept: the software layers are complete and measured, the physical network beneath the inventories is simulated.

- **A 118-node executable BPMN model across five lanes**, run by Kogito and compiled to Java at build time, so a modelling error breaks the build rather than a customer order.
- **17 TM Forum Open APIs (v4)** served by nine Quarkus microservices, each with its own PostgreSQL database. 13 were run against the official Conformance Test Kits, the remaining 4 against their OpenAPI schema.
- **52 Drools business rules** lifted out of the code, each file owned by the service that consults it, so changing a discount or a threshold requires no code change.
- **Orchestrated compensation (saga)**: when a journey fails, four handlers undo it in reverse order, the fibre port returns to the pool and the customer is told why.
- **Event-driven choreography on Apache Kafka**, 23 topics in CloudEvents 1.0, with instance correlation to wake a waiting process.
- **Real billing** through Kill Bill behind an anti-corruption layer.
- **Angular portal** with four role areas, plus a read-only customer assistant built on a language model where every figure comes from the Open APIs.

Measured on the running platform: 117 unit tests, 10 end-to-end journeys with 54 assertions, 50 concurrent orders with no drift in the median response time.

A defect found and fixed through measurement: under concurrency, port reservation handed three ports to ten orders, one of them to eight, while every call returned HTTP 200. The remedy is a conditional write evaluated by the database, because an application-level lock does not protect state that two requests share.

Java 17 • Quarkus • Kogito • Drools • Apache Kafka • PostgreSQL • Hibernate Reactive • Kill Bill • Angular • Docker • GitHub Actions

Deep Learning Engineer, Computer Vision — Smart Elastic Solutions

June – July 2025

SSD MobileNet object detection for an industrial use case. Purpose-built dataset, model tuning and an automated Python pipeline.

Front-End Developer — AllaCarta

July 2024

Integration of a React template into a production application.

Full-Stack Developer — Archicompass (French startup)

February – September 2023

Health-compliance application in Spring Boot and Angular. Weekly Scrum sprints, authentication and authorisation through Keycloak.

Web Developer — ADS Valley

June – July 2022

Application for sharing internship experiences.

EDUCATION

Engineering Degree in Software Engineering (five-year programme, equivalent to an MSc)

Higher Institute of Computer Science, University of Tunis El Manar • 2023 – 2026

MSc by Research, Artificial Intelligence — coursework and examinations completed, thesis in progress

Higher Institute of Computer Science, University of Tunis El Manar • 2024 –

BSc, Software Engineering and Information Systems

Higher Institute of Computer Science, University of Tunis El Manar • 2020 – 2023

CERTIFICATION

Introduction to TM Forum and the Open Digital Architecture — TM Forum, May 2026

SKILLS

Process and rules — BPMN 2.0, Kogito, Drools, compensation and sagas

Telecom — TM Forum Open APIs v4, eTOM, SID, ODA, Order-to-Activate

Services — Java 17, Quarkus, Spring Boot, REST, Hibernate Reactive

Event-driven — Apache Kafka, CloudEvents, idempotence, correlation

Data — PostgreSQL, data modelling, concurrent writes

Front end — Angular, TypeScript

Engineering practice — Docker, Docker Compose, GitHub Actions, load testing